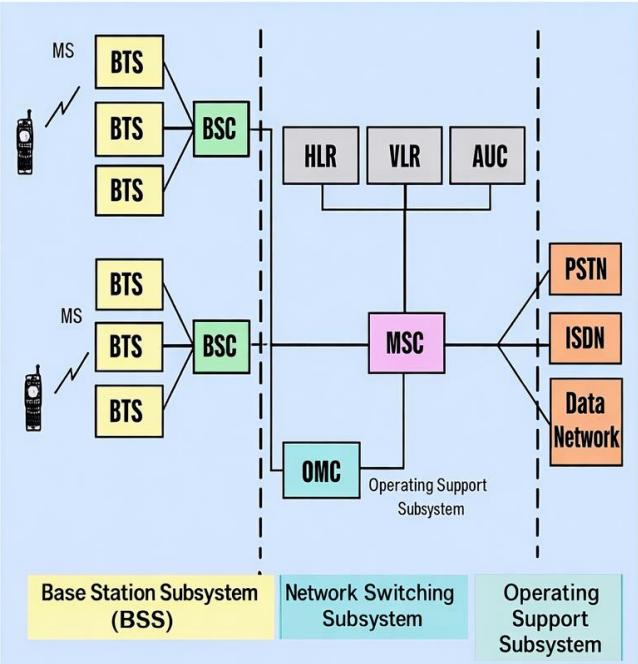
GSM Architecture

GSM SubSystems

GSM architecture is mainly divided into three Subsystems

1. **Base Station Subsystem (BSS)**
2. **Network & Switching Subsystem (NSS)**
3. **Operations & Support Subsystem (OSS)**

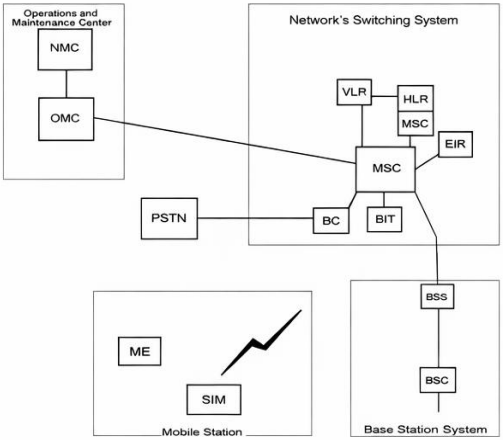
Mobile Station ***sometimes*** included in BSS



(Figure: GSM System Architecture)

- **MS** = Mobile Station
- **BTS** = Base Transceiver Station
- **BSC** = Base Station Controller
- **MSC** = Mobile Switching Center
- **VLR** = Visitor Location Register
- **HLR** = Home Location Register
- **EIR** = Equipment Identity Register
- **ISCN/PDN** = Integrated Services Communication Network / Public Data Network
- **PSTN** = Public Switched Telephone Network
- **AUC** = Authentication Center
- **Um**: Mobile Station ↔ Base Transceiver Station
- **A-bis**: Base Transceiver Station ↔ Base Station Controller
- **A**: Base Station Controller ↔ Mobile Switching Center
- **B**: Mobile Switching Center ↔ Visitor Location Register
- **C**: Mobile Switching Center ↔ Home Location Register
- **D**: Mobile Switching Center ↔ Equipment Identity Register
- **E**: Mobile Switching Center ↔ Integrated Services Communication Network / Public Data Network
- **F**: Mobile Switching Center ↔ Public Switched Telephone Network
- **G**: Home Location Register ↔ Authentication Center

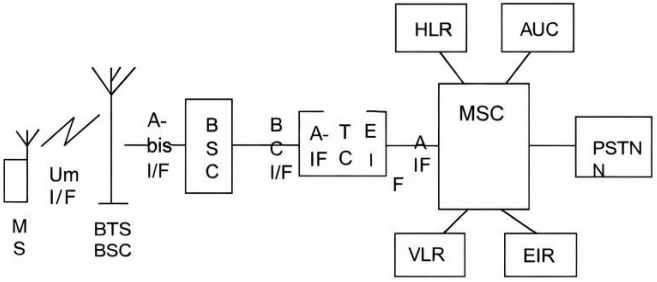
GSM Network Architecture

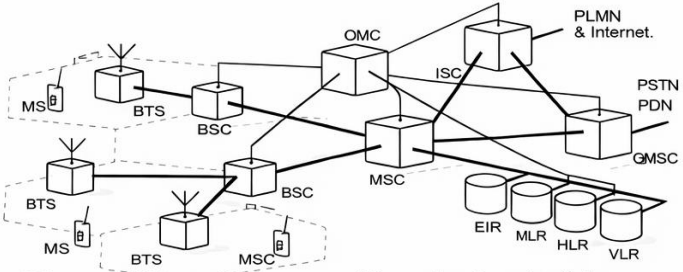


Interface/Connection

Name	Interface
Um	MS <=> BTS
A-bis	BTS <=> BSC
A	BSC <=> MSC
B	MSC <=> VLR
C	MSC <=> HLR
D	MSC <=> EIR
E	MSC <=> ISCN/PDN
F	MSC <=> PSTN
G	HLR <=> AUC

Simplified GSM Architecture





BTS	Base Transceiver Station
BTS	Base Station Controller
BSC	Mobile Switching Center
MSC	Operation ʌsslt
OMC	International Switching Center

MS	Base Transceiver Station
HLR	Mobile Switching Center
VLR	Visitor Location Register
EIR	Equipment Idenity Register
/SC	Rase Station Center
TMC	Operation and Maintenance Center

Mobile Station (MS)

The MS consists of two parts

1.Mobile Equipment (ME)

2.Subscriber Identity module (SIM)

Mobile Equipment (ME)

The ME is the only part of the GSM network which the subscriber will really see.

There are three main types of ME, these are listed below:

1. **Vehicle Mounted**
2. **Portable Mobile Unit**
3. **Hand portable Unit**

FULL SIZE SIM CARD

Subscriber Identity module (SIM)

The SIM is a card which plugs into the ME.

This card identifies the MS subscriber and also provides other information regarding the services that subscriber should receive.

The SIM card, and the high degree of inbuilt system security, provides protection of the subscriber's information and protection of networks against *fraudulent* access.

The SIM can be protected by use of Personal Identity Number (PIN) password, similar to bank/credit charge cards, to prevent unauthorized use of the card.

SIM cards are designed to be difficult to duplicate.

By making a distinction between the ubscriber identity and the ME identity, GSM can route calls and perform billing based on the identity of the 'subscriber' rather than the equipment or its location.

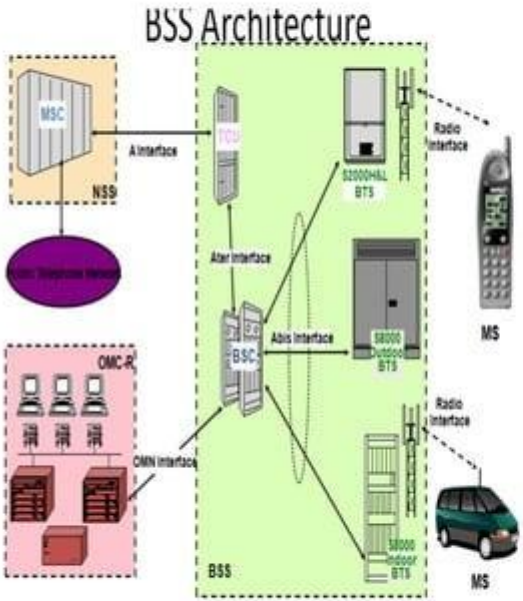
The SIM contains several pieces of information:

1. **International Mobile Subscriber Identity (IMSI)**
2. **Temporary Mobile Subscriber Identity (TMSI)**
3. **Location Area Identity (LAI)**
4. **Subscriber Authentication Key (Ki)**
5. **Mobile Station Integrated Services Digital Network (MSISDN)**

The SIM is capable of storing additional information such as accumulated call charges.

The SIM also executes the Authentication Algorithm

Base Station Controller (BSC)



the BSC provides the control for the BSS. Any operational information required by the BTS will be received via the BSC.
Likewise any information required about the BTS (by the OMC for example) will be obtained by the BSC.
The BSC incorporates a digital switching matrix, which it uses to connect the radio channels on the air interface with the terrestrial circuits from the MSC.
The BSC switching matrix also allows the BSC to perform "handovers" between radio channels on BTSs, under its control, without involving the MSC

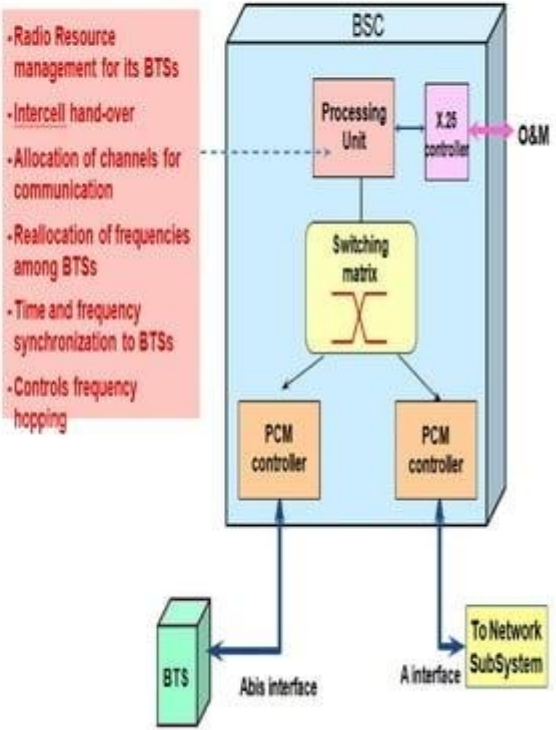
Base Station Controller (BSC)

- BSC manages the base stations over the "A-bis" interface
  - *The A-bis link can be copper, fiber, or microwave*
- The BSC handles:
  - *Handoffs*
  - *Cell rankings*
  - *Locating*
  - *Power control*
  - *Channel allocation*
  - *Frequency/code allocations*
  - *Coding (if not in the BTS)*
  - *Limited switching*
- Some vendors have BSC functionality in the MSC

Ericsson CDMA BSC

Samsung CDMA BSC

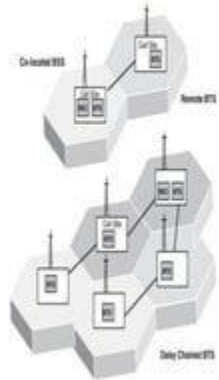
BSC General Architecture and Functions



Base Transceiver Station – BTS

The BTS provides the air interface connection with the MS.
It also has a limited amount of Control functionality which reduces the amount of traffic passing between the BTS and BSC.
Where the BSC and BTS are both shown to control a function, the control is divided between the two, or may be located wholly at one.

BSS Functionality	Control
Terrestrial Channel Management	
Channel Allocation	BSC
Radio Channel Management	BSC
Channel Configuration Management	BSC
Handover Control	BSC
Frequency Hopping	BSC/BTS
Traffic Channel Management	BSC/BTS
Control Channel Management	BSC/BTS
Encryption	BSC/BTS
Paging	BSC/BTS
Power Control	BSC/BTS
Channel Coding/Decoding	BTS
Timing Advance	BTS
Idle Channel Observation	BTS
Measurement Reporting	BTS

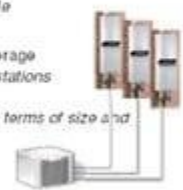


BSS Configurations

- The maximum number of BTSs which may be controlled by one BSC is not specified by GSM.
- The BTSs and BSC may either be located at the same cell site "co-located", or located at different sites "Remote".
- Another BSS configuration is the daisy chain.
- Problem- transmission delay through the chain.

Micro and Pico BTSs Becoming Common

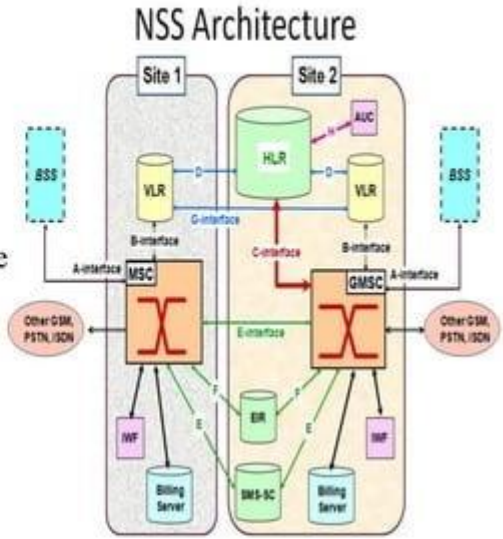
- Micro base stations have lower power and lower capacity
  - *Typically 20-30 dBm supporting 8-20 channels*
  - *Purpose is to relieve capacity in hot spots*
  - *All features of the technology are typically available*
  - *Must look small and aesthetically pleasing*
- Pico base stations are necessary for good indoor coverage
  - *Even smaller and more discrete than micro base stations*
  - *Typically 10-20 dBm supporting 8-20 channels*
  - *Beginning to approach a 'cordless' base station in terms of size and power*

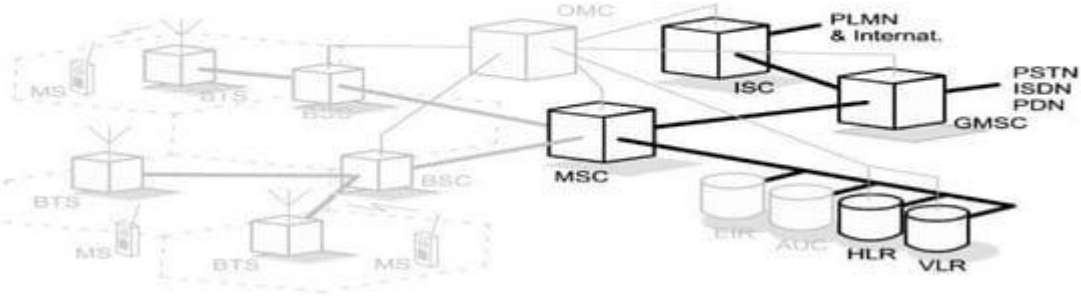


Network Switching System (NSS)

The Network Switching System includes the main switching functions of the GSM network.
It also contains the databases required for subscriber data and mobility management.
The components of the Network Switching System are listed below:

1. Mobile Services Switching Centre – MSC
2. Home Location Register – HLR
3. Visitor Location Register – VLR
4. Equipment Identity Register – EIR
5. Authentication Centre – AUC
6. Interworking Function – IWF
7. Echo Canceller – EC





Mobile Switching Centre (MSC)

- MSC is the ***heart*** of the system, controlling the Switching & Billing.
- The MSC can carry out different functions depending upon its position in the network.
- When provides interface between PSTN & BSS in GSM network then known as a *Gateway MSC*
- provides service to MSs located within a *defined* geographic coverage area.
- The network typically contains more than *one* MSC.
- One MSC is capable of supporting a regional capital with approximately *one million* inhabitants

Call Processing

1. control of data/voice call setup
2. inter-BSS and inter-MSC handovers
3. control of mobility management (subscriber validation and location).

Operations and Maintenance Support

1. database management
2. traffic metering and measurement
3. A Man–machine interface.

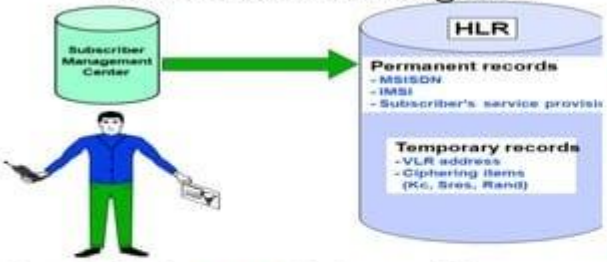
Internetwork Interworking

1. Interface between the GSM network and the PSTN.

Billing

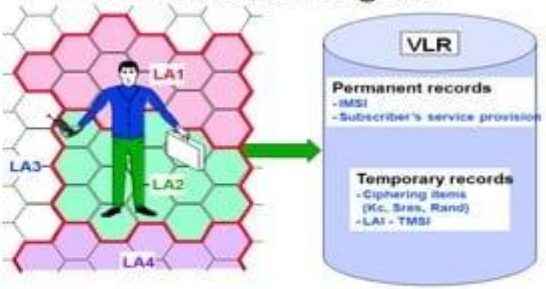
1. Collects call billing data.

Home Location Register



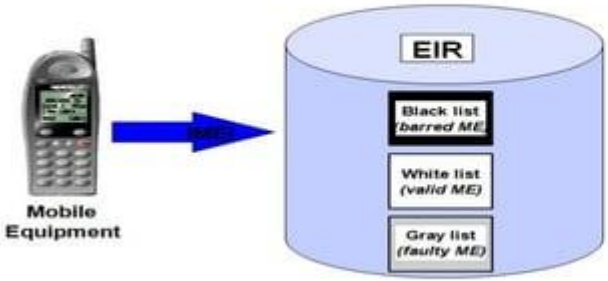
- The HLR is the master database which contains each user's service profile.
- Various identification numbers and addresses are stored, as well as authentication parameters.
- The data it contains is remotely accessed by all the MSCs and the VLRs in the network.
- Although the network may contain more than one HLR, there is only one database record per subscriber .
- The subscriber data may be accessed by either the IMSI or the MSISDN number.

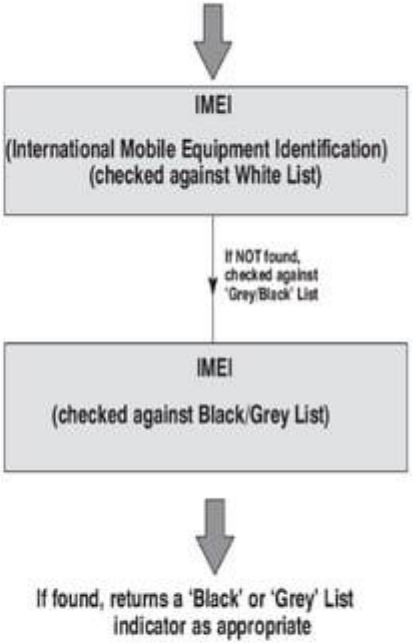
Visitor Location Register



- VLR is a **temporary** database for all user currently located in the system including roamers & non-roamers.
- The data exists for only as long as the subscriber is *"active"* in the ***particular area*** covered by the VLR.
- The VLR database will therefore contain some ***duplicate*** data as well as ***more precise*** data relevant to the subscriber.
- This function eliminates the need for ***excessive*** and ***time-consuming*** references to the "home" HLR database.

Equipment Identity Register

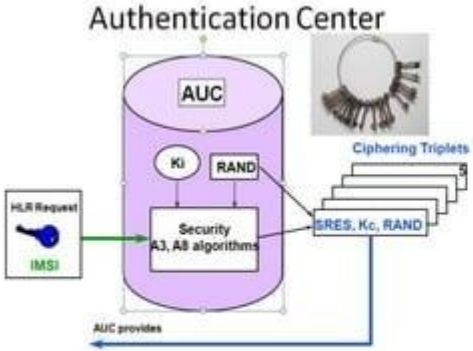




Equipment Identity Register (EIR)

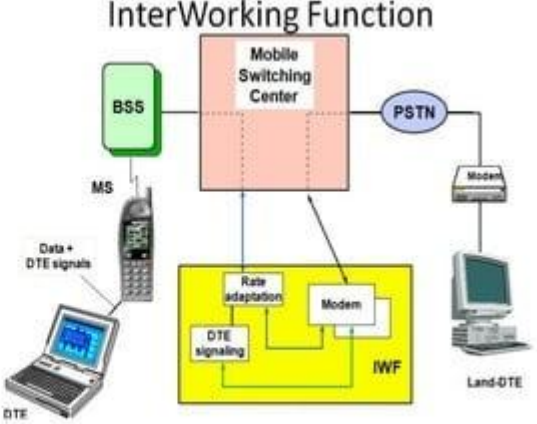
The EIR contains a **centralized** database for validating the IMEI.
This database is concerned solely with MS equipment and not with the subscriber who is using it to make or receive a call.
The EIR database consists of lists of IMEIs (or ranges of IMEIs) organized as follows:

1. **White List**
2. **Black List**
3. **Grey List**



Authentication Centre (AuC)

The AuC is a ***processor*** system that performs the "authentication" function. It is normally ***co-located*** with the HLR as it will be required to continuously access and update, as necessary, the system subscriber records.
The authentication process will usually take place each time the subscriber "initializes" on.

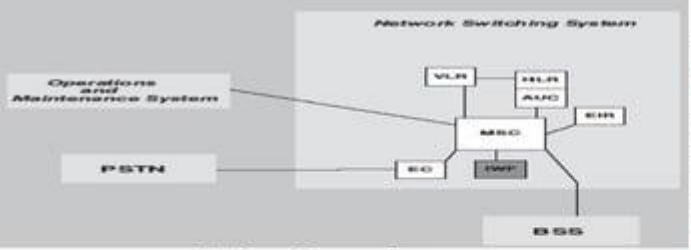


Interworking Function (IWF)

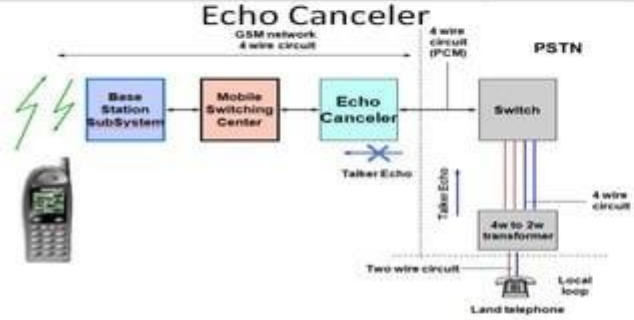
IWF provides the function to enable the GSM system to interface with the various forms of public and private data networks.
The basic features of the IWF are

1. **Data rate adaptation.**
2. **Protocol conversion.**

Some systems require more IWF capability than others, this depends upon the network to which it is being connected.

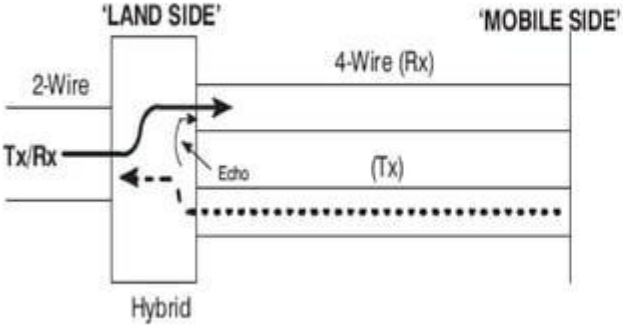


The IWF also incorporates a "modem bank", which may be used when for example, the GSM **Data Terminal Equipment (DTE)** exchanges data with a land DTE connected via an analogue modem.



Echo Canceller (EC)

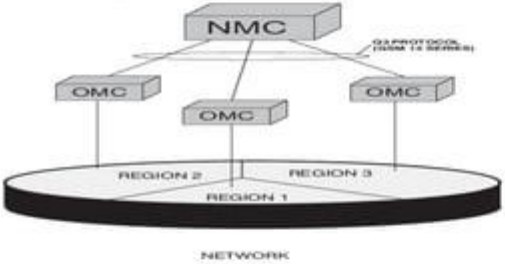
An EC is used on the PSTN side of the MSC for all voice circuits.
Echo control is required at the switch because the inherent GSM system delay(180 ms approx) can cause an unacceptable echo condition.
This would not be apparent to the MS subscriber, but for the inclusion of a 2-wire to 4-wire hybrid transformer in the circuit.
The transformer causes the echo. This does not affect the land subscriber.



Operations & Support System (OSS)

The OSS provides the capability to ***manage*** the GSM network remotely.
This area of the GSM network is not currently ***tightly*** specified by the GSM specifications.
It is left to the network provider to decide what ***capabilities*** they wish it to have.
The Operations and Maintenance System comprises of two parts:

1. **Network Management Centre (NMC)**
2. **Operations and Maintenance Centre (OMC)**



OMC (REGIONAL)	NMC (GLOBAL)
Multiple OMCs per network	Single NMC per network
Regionalized network management	Global network management
Employed in daily operations	Employed in long term planning
Used by network operators	Used by network managers and planners
	24 hour supervision

Functionality of the NMC
Monitors nodes on the network
Monitors GSM network element statistics
Monitors OMC regions & provides information to OMC staff
Passes on statistical information from one OMC region to another to improve problem solving strategies
Enables long term planning for the entire network

Operations and Maintenance Centre (OMC)

- The OMC provides a central point from which to control and monitor the other network entities (i.e. base stations, switches, database, etc).
- It also monitors the quality of service being provided by the network.
- There are two types of OMC these are:

1.OMC-R

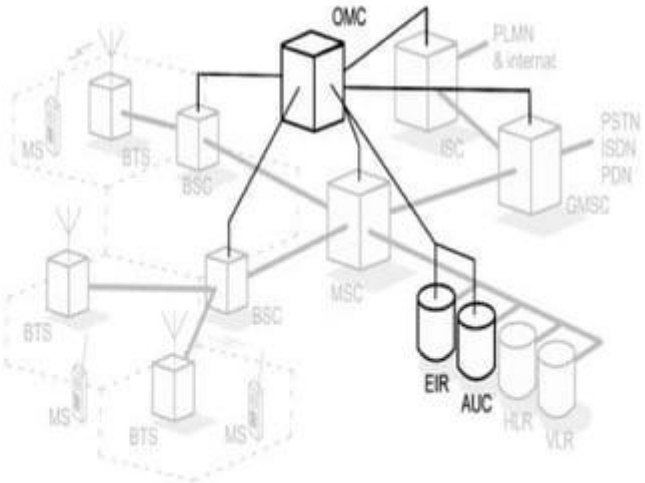
OMC controls specifically the Base Station System.

2. OMC-S

OMC controls specifically the Network Switching System.

The OMC should support the following functions as per ITS–TS recommendations:

1. **Event/Alarm Management.**
2. **Fault Management.**
3. **Performance Management.**
4. **Configuration Management.**
5. **Security Management.**



GSM Channel are divided into two types:

Traffic channels (TCHs):

The traffic channes are intended to carry encoded speech or user data.

Traffic channels are intended to carry encoded speech and user data.

Full rate traffic channels at a net bit rate of 22.8 Kb/s (TCH/F)

Half rate traffic channels at a net bit rate of 11.4 Kb/s (TCH/H)

Speech channels are defined for both full rate and half rate traffic channels.

Data channels support a variety of data rates (2.4, 4.8 and 9.6 Kb/s) on both half and full rate traffic channels. The 9.6 Kb/s data rate is only for full rate application

The traffic channels(TCH) support two types of information rates Full rate (TCH/F) and Half rate (TCH/H) When transmitted as full rate, the user data is occupied within TS per frame. When transmitted as half rate, the user data is occupied into the same time slot but sent in alternate frames.

The 26th frame contains idle bits if full rate TCHs are used and contains SACCH data if half rate TCHS are used

- Full Rate TCH for data and speech channels:-

A. **Full - rate Speech Channel)TCH/Fs):** At 16 kbps the full rate speech channel is digitized. The full rate speech channel caries 55.8kbps after adding the GSM channel coding to the digitized speech.

B. **Full-rate Data Channel for 9600 bps (TCH/F9.6):** The full rate traffic data channel contains raw data that is transmitted at 9.6 kbps. After the application of additional forward error correction coding with the GSM standards, 9600 kbps is transferred at 22.8 kbps.

C. **Full-rate Data Channel for 4500 bps(TCH/F4.8):** The full rate traffic date channel contains data that is transmitted at 4.8 Kbps. After the application of additional forward error correction coding with GSM standards, the 4.8 kbps is transferred at 22.8 kbps.

D. **Full Rate Data Channel for 2400 bps (TCH/F2.4):** The full rate traffic data channel contains raw data that is transmitted at 2.4 kbps. After the application of additional forward error correction coding with GSM standards, the 2.4 kbps data is transferred at 22.8 kbps.

Half Rate TCH for data and speech channels:

A. **Half Rate Speech Channels (TCH/HS):** The half tate speech channel can carry digitized speech that is sampled at a rate half that of full rate channel. GSM anticipates the availibility of speech coders. It can digitize speech at about 6.5 kbps. After adding GSM channel coding to the digitized speech, the half rate Speech channel will carry 11.4 kbps.

B. **Half Rate Data Channel for 4800 bps (TCH/H4.8):** The half rate traffic data channel carries raw data that is sentat 4800 bps. After the application of forward error correction using GSM standards, 4800 bps data is sent at 11.4 kbps.

C **Half Rate Data Channe for 2400 kbps (TCH/H 2.4):** The half rate traffic data channel carries raw user data that is sent at 2400 bps. After application of additional forward error corretion using GSM standards, 2400 bps data is sent to 11.4 bps.

Control Channel (CCH):

. Control channels carry signaling information between an MS and a BTS.

a) **Broadcast control channel:**

. Broadcast control channels are transmitted in downlink direction only i.e. only transmitted by BTS.

. The broadcast channels are used to broadcast synchronization and general network information to all the MSs within a cell.

Signaling Channels are divided into three types

a. **FREQUENCY CORRECTION CHANNEL (FCCH):**

. Used for the frequency correction / synchronization of a mobile station.

. The repeated (every 10 sec) transmission of Frequency Bursts is called FCCH.

b. **SYNCHRONISATION CHANNEL (SCH):**

. Allows the mobile station to synchronize time wise with the BTS.

. Repeated broadcast (every 10 frames) of Synchronization Bursts is called (SCH).

c. **BROADCAST CONTROL CHANNEL (BCH):**

. The BROADCAST CONTROL CHANNEL (BCCH) is used to broadcast control information to every MS within a cell.

MULTIPLE ACCESS TECHNIQUES:

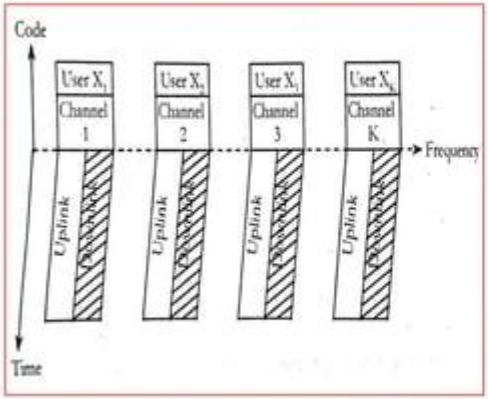
- Sometimes a satellite's service is present at a particular location on the earth station and sometimes it is not present. That means, a satellite may have different service stations of its own located at different places on the earth. They send carrier signal for the satellite.
- In this situation, we do multiple access to enable satellite to take or give signals from different stations at time without any interference between them.
- Following are the **three types** of multiple access techniques.

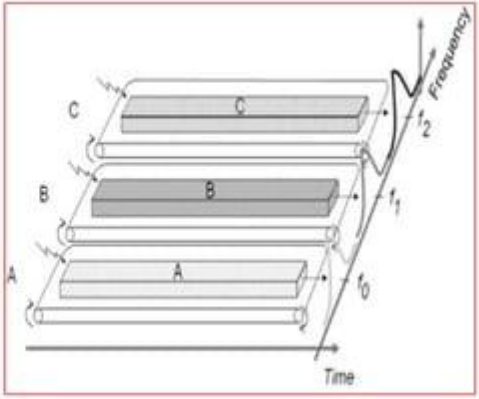
1. FDMA (Frequency Division Multiple Access)
2. TDMA (Time Division Multiple Access)
3. CDMA (Code Division Multiple Access)
4. SDMA (Space Division Multiple Access)

FDMA (Frequency Division Multiple Access)

- Frequency Division Multiple Access (FDMA) is one of the most common analogue multiple access methods.

- The frequency band is divided into channels of equal bandwidth so that each conversation is carried on a different frequency
- In FDMA method, guard bands are used between the adjacent signal spectra to minimize crosstalk between the channels. A specific frequency band is given to one person, and it will received by identifying each of the frequency on the receiving end. It is often used in the first generation of analog mobile phone.





Features of FDMA

1) In FDMA system during the period of the call, no other user can share the same channel, so channel carries only one phone circuit at a time.

2) This system have big disadvantages, while radio sations broadcast 24 hours a day, mobile communication typically takes place for only a few minutes at a time. Assigning a separate frequency for each possible communication scenario would be a tremendous waste of frequency resources.

3) The fixed assignment of a frequency to a sender makes the scheme very inflexible and limits the number of senders.

4) After the assignment of a voice channel, the base station and the mobile transmit simultaneously and continuously.

5) Because in FDMA each channel supports only one circuit per carrier so bandwidth is very low near about 30 kHz in 1G system.

6) Designing of FDMA systems is easy compared with TDMA systems, now a days for improving the TDMA system digital signal processing technique is used.

7) The FDMA mobile unit uses duplexers since both the transmitter and receiver operate at the same time. This results in an increase in the cost of FDMA subscriber units and base stations.

8) High capacity and accurate filter are required to minimize adjacent channel interferences.

9) Due to continuous transmission scheme, fewer bits are required for overhead purposes like bits for framing and synchronization as compared to TDMA.

The number of channels supported by FDMA System are ,

$$N = \frac{(B_t - 2B_g)}{B_c}$$

Where,

B_t = Total spectrum allocation.

B_g = Guard band allocated at the edge of the allocated spectrum band.

B_c = Channel bandwidth.

Advantages

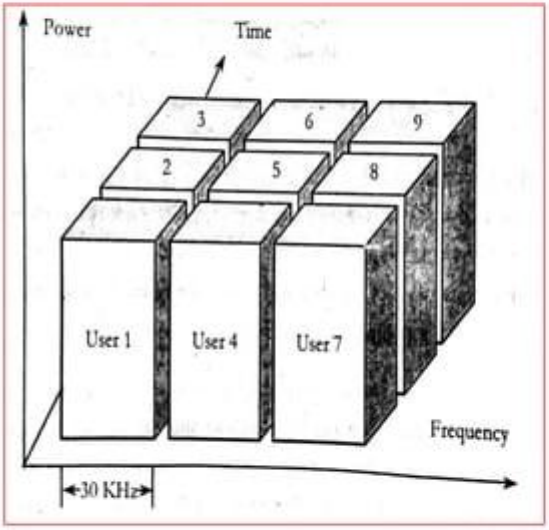
1) Very Simple to design.
2) Narrowband (no Inter Symbol Interference).
3) Synchronization is easy.
4) No interference among users in a cell.

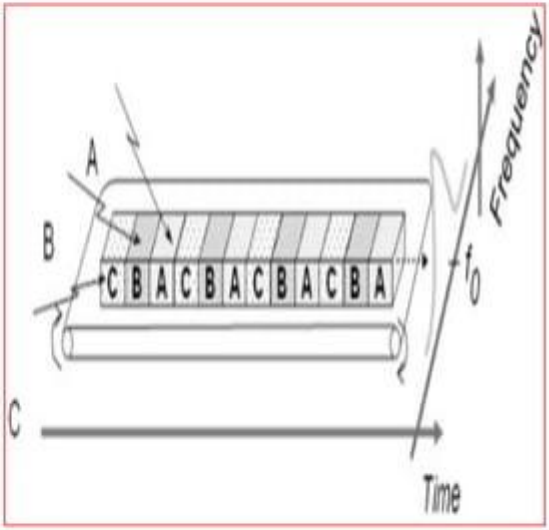
Disadvantages

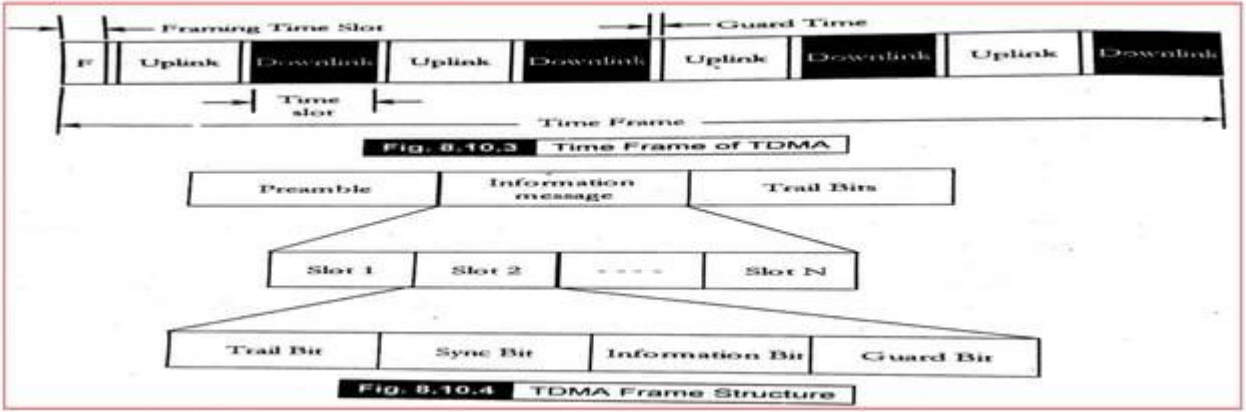
1) Narrowband interfernce.
2) Static spectrum allocation.
3) Frequency reuse is a problem.
4) High Q analog filters.
5) Large guard band required.

TDMA (Time Division Multiple Access) :

- Time Division Multiple Access (TDMA) is a digital cellular telephone communication technology.
- It facilitates many users to share the same frequency without interference.
- Its technology divides a signal into different timeslots, and increases the data carrying capacity.







The features of TDMA include the following,

1) TDMA shares the single carrier frequency with several users, where each user makes use of non-overlapping timeslots.

2) Data Transmission for user of TDMA system is discrete bursts. The result is low battery consumption, since the subscriber can be turned off when not in use.

3) Because of discontinuous transmission in TDMA, the handoff process is simpler, since it is able to listen for other base stations during idle time slots.

4) Equalization is required, since transmission rates are generally very high as compared to FDMA channels.

Some of the advantages and disadvantages of TDMA are,

Advantages

1) Better suited for digital.
2) Often gets higher capacity (3 times higher here).
3) Relaxes need for high Q filters.

Disadvantages

1) Strict synchronization and guard time needed.
2) Still susceptible to jamming, other cell interference.
3) Often requires equalizer.

CDMA (Code Division Multiple Access):

- Code Division Multiple Access system is very different from time and frequency multiplexing.
- In this system, a user has access to the whole bandwidth for the entire duration.
- The basic principle is that different codes are used to distinguish among the different users.
- A code called **spreading code** is used to perform this action. Each code is orthogonal to all other code words.

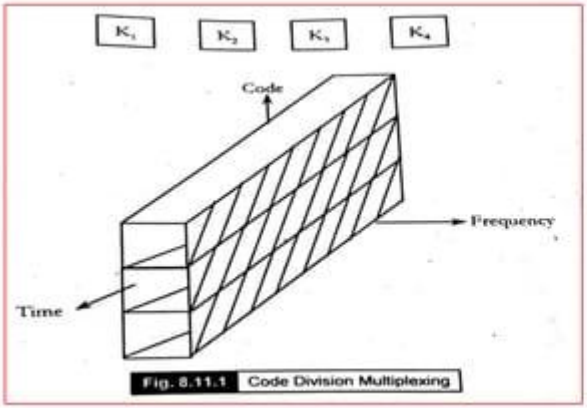
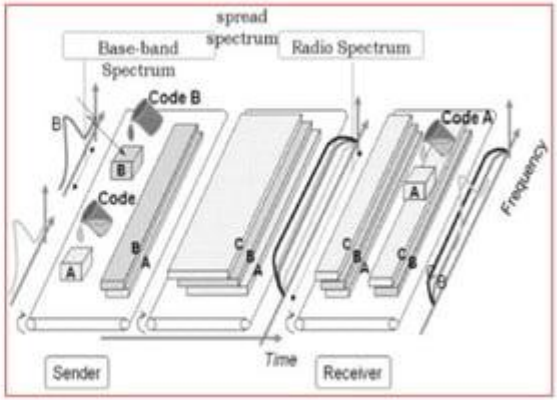


Fig. 8.11.1 Code Division Multiplexing



Advantages of CDMA:

- CDMA has a soft capacity.

The greater the number of codes, the more the number of users. All signals must have more or less equal power at the receiver.

- Rake receivers can be used to improve signal reception.
- Flexible transfer may be used. Mobile base stations can switch without changing operator. Two base stations receive mobile signal and the mobile receives signals from the two base stations.
- Transmission Burst – reduces interference.

Disadvantages of CDMA:

The disadvantages of using CDMA are as follows −

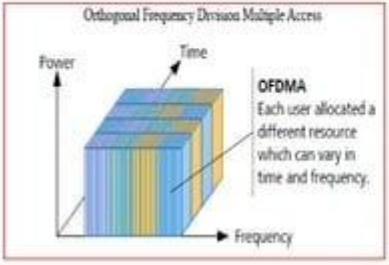
- The code length must be carefully selected. A large code length can induce delay or may cause interference.
- Time synchronization is required.
- Gradual transfer increases the use of radio resources and may reduce capacity.
- As the sum of the power received and transmitted from a base station needs constant tight power control. This can result in several handovers.

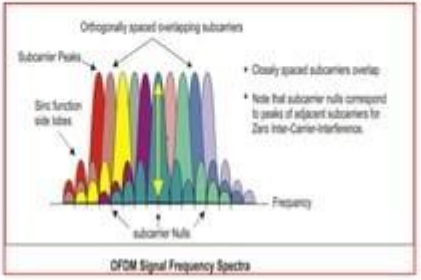
SDMA (Space Division Multiple Access):

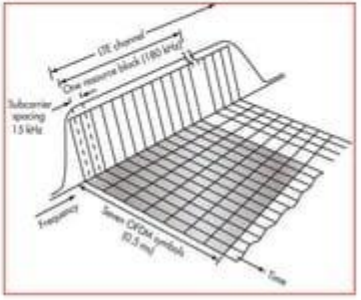
- Spatial division multiple access (SDMA) is a channel access method used in mobile communication systems which reuses the same set of cell phone frequencies in a given service area.
- Two cells or two small regions can make use of the same set of frequencies if they are separated by an allowable distance (called the reuse distance).
- The main advantage of SDMA is frequency reuse.
- SDMA increases the capacity of the system and transmission quality by focusing the signal into narrow transmission beams.
- Through the use of smart antennas with beams pointed at the direction of the mobile station, SDMA serves different Users within the same region.
- Provided the reuse distance is preserved in the network architecture, interference can be near zero, even if mobile stations use the same allocated frequencies.

Orthogonal frequency-division multiple access (OFDMA):

- OFDMA is the access technique used in Long-Term Evolution (LTE) cellular systems to accommodate multiple users in a given bandwidth.
- Orthogonal frequency division multiplexing (OFDM) is a modulation method that divides a channel into multiple narrow orthogonal bands that are spaced so they don't interfere with one another.
- Each band is divided into hundreds or even thousands of 15-kHz wide subcarriers.
- OFDMA is an updated version of frequency-division multiplexing (FDM) technology used to divide packets of information into separate bands that are carried by separate signals.
- This form of communication is an upgrade that parallels the switch of internet carriers to Wi-Fi 6 wireless, as well as the upgrade of phone carriers to 4G and 5G LTE.
- Instead of the traditional analog modulation used in multiplexing, OFDMA uses carrier signal waves, called subcarriers, to move small bits of information in a more streamlined fashion.







Advantages and disadvantages of OFDMA

Implementing OFDMA can provide the following advantages:

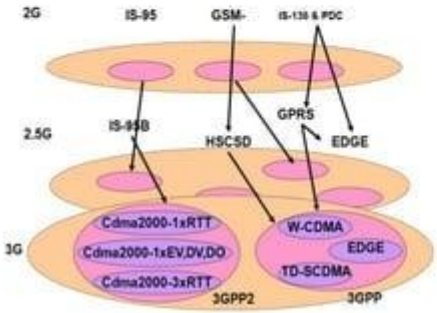
- Higher diversity and efficiency of frequency
- Provides lower interference between cells.
- More flexibility as channels and sub-channels can be toggled on and off.
- Better coverage over networks.

Disadvantages include:

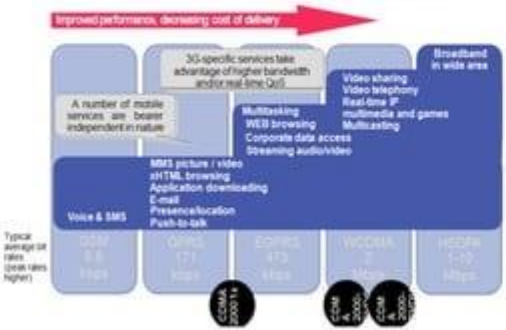
- The diversity of frequencies is conditional on how subcarriers are assigned to users, and can thus become very complex.
- Requires extra power because it is always on and ready to send a transmission.
- Has a higher sensitivity than other channel types

•3G, 3.5G AND 4G (LTE)

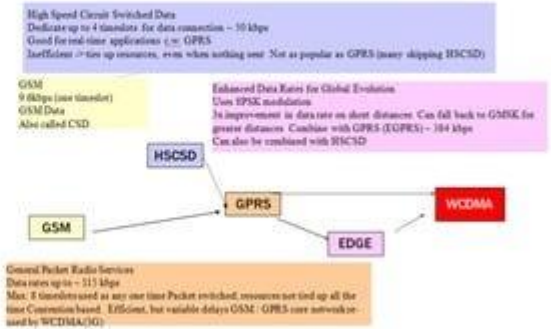
EVOLUTION FROM 2G



SERVICE ROADMAP



GSM EVOLUTION TO 3G



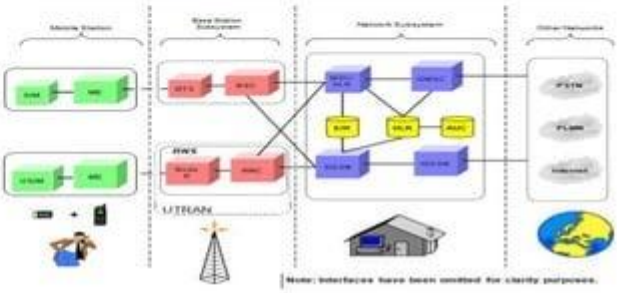
3G OVERVIEW

- 3G is created by ITU-T and is called IMT-2000

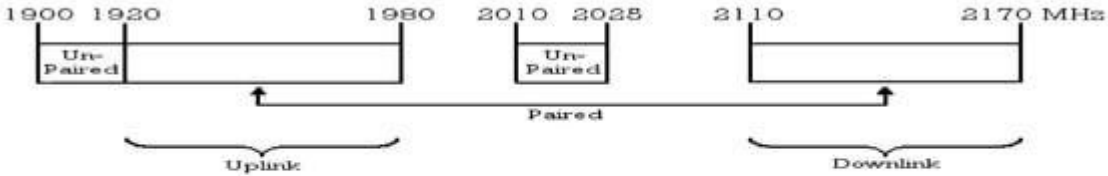
Universal Mobile Telecommunications System (UMTS)

- UMTS is an upgrade from GSM via GPRS or EDGE
- The standardization work for UMTS is carried out by Third Generation Partnership Project (3GPP)
- Data rates of UMTS are:
  - 144 kbps for rural
  - 384 kbps for urban outdoor
  - 2048 kbps for indoor and low range outdoor
- Virtual Home Environment (VHE

UMTS ARCHITECTURE



UMTS FREQUENCY SPECTRUM



UMTS Band

- 1900-2025 MHz and 2110-2200 MHz for 3G transmission
- In the US, 1710–1755 MHz and 2110–2155 MHz will be used instead, as the 1900 MHz band was already used.

UMTS network architecture consists of three domains

•Core Network (CN): Provide switching, routing and transit for user traffic

•UMTS Terrestrial Radio Access Network (UTRAN): Provides the air interface access method for user equipment.

•User Equipment (UE):Terminals work as air interface counterpart for base stations.The various identities are: IMSI,TMSI, P-TMSI, TLLI, MSISDN, IMEI, IMEISV

Introduction to 3G System:

•The third generation of mobile telecommunications technology is 3G.

• An information transfer rate of at least 200 kbit/s is provided by the services which are supported by 3G network.

• 3G Systems give us with mobile broadband access to smart phones, mobile modems, Speed of up to 2 Mbps.

• Then the allocation of communication spectrum between 400 MHz to 3 GHz for 3G was done.

•3G standard was then approved by both the government & communications companies.

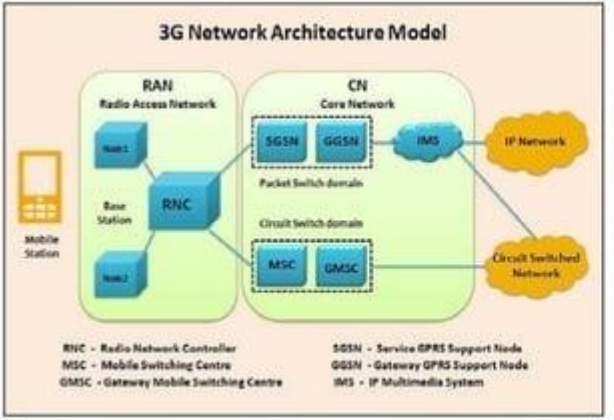
• In 1998, NTT DoCoMo in Japan launched the first pre-commercial 3G network, branded as FOMA.

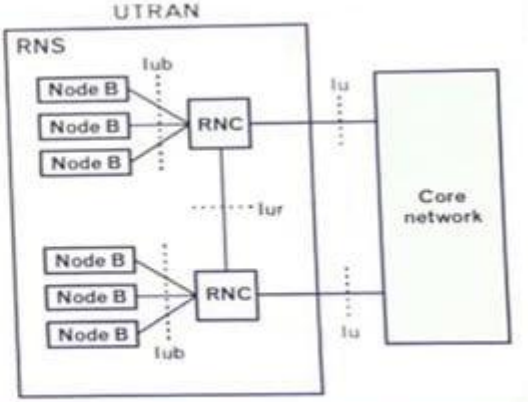
3.5G High Speed Packet Access (HSPA)

High Speed Packet Access (HSPA) is an amalgamation of two mobile telephony protocols, High Speed Downlink Packet Access (HSDPA) and High Speed Uplink Packet Access (HSUPA), that extends and improves the performance of existing *WCDMA* protocols

5.G introduces many new features that will enhance the UMTS technology in future. 1xEV-DV already supports most of the features that will be provided in 3.5G.These include:

- Adaptive Modulation and Coding
- Fast Scheduling
- Backward compatibility with 3G
- Enhanced Air Interface





Architecture of 3G cellular systems.

- The 3G system consists of two main parts: the User Equipment (UE) and the UMTS Terrestrial Radio Access Network (UTRAN). The UE is the mobile phone and the UTRAN is the base station and the network intelligence.
- The UTRAN is the new radio access network designed especially for UMTS.
- The UTRAN consists of radio network controllers (RNCs) and Node Bs (base stations).
- Together, these entities form a radio network subsystem (RNS). See Figure 8.4.
- The internal interfaces of the UTRAN include the Iub and Iur.
- The Iub connects a Node B to the RNC and the Iur is a link between two RNCs.

Node B

- Node B is the UMTS equivalent of a base station transceiver.
- Functions that are performed by a Node B include the followin'lowing:
  - Node B logical O&M implementation;
  - Mapping of Node B logical resources to the hardware resources;
  - Transmitting of system information messages according to istntrstruction given by the RNC;
  - Macrodiversity combining/splitting of data streams transferred over several Node Bs;
  - Transmission power control (in FDD mode);
  - Reporting of uplink interference measurements and DL power information.

Radio Network Controller (RNC)

An RNC is comparable to a BSC in GSM networks. Functions that are performed by the RNC include the following:

- Control of Node B logical operation and maintenance (O&M) resources;
- Transmission of common system information messages;
- Macrodiversity combining/splitting of data streams transsferred over several Node Bs;
- Soft handovers, including intra., inter-, and soft handovers;
- Allocation of radio resources to users;
- Assignment of DL channelization codes;
- Uplink outer loop power control;
- Handover control;
- Macrodiversity combinining of data streams transferred over several cell channels;
- Reporting and interworking with the core network of different operators.

Features of 3G Cellular Systems

- Speed of up to 2 Mbps;
- Increased bandwidth and datata transfer rates;
- Support for large email and messages;
- Enhanced audio capabilities;
- Wide spectrum to relieve overerovcrowding in existing systems;
- More bandwidth, security, and reliability,
- Better compatibility between service providers in fall NodeBs logical resources.

4G System:

•The main difference between 3G and 4G is the data rate.

• The key technologies that have made 4G possible are MIMO (Multiple Input Multiple Output) and OFDM (Orthogonal Frequency Division Multiplexing).

• The most important 4G standards are WiMAX and LTE.

•The fourth generation of mobile telecommunication technology standards is 4G.
It is basically a successor of third generation (3G) standards.

•4G systems provide mobile ultra broadband Internet access, example, to Smartphone's & other mobile devices.

• A packet-switched network is provided in 4G systems instead of the circuit-switched infrastructure

4G (LTE)

•LTE stands for Long Term Evolution

•Next Generation mobile broadband technolog

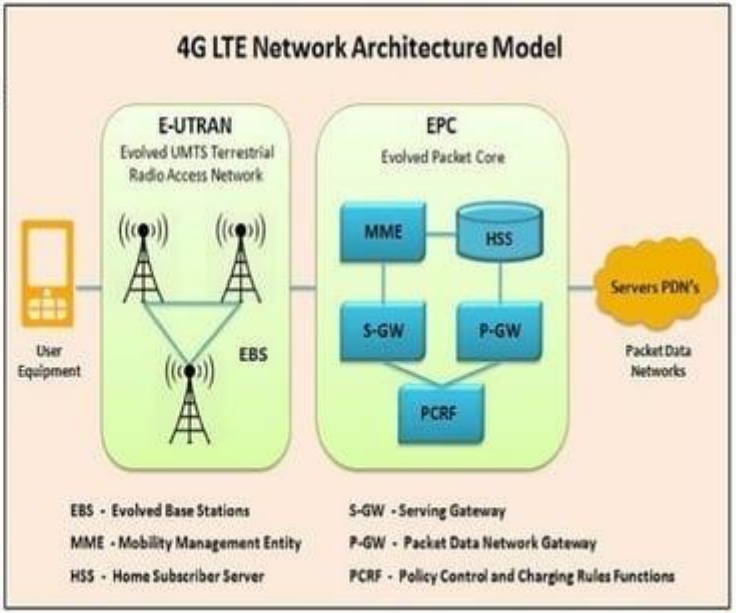
•Promises data transfer rates of 100 Mbps

•Based on UMTS 3G technology

•Optimized for All-IP traffic

•ADVANTAGES OF LTE

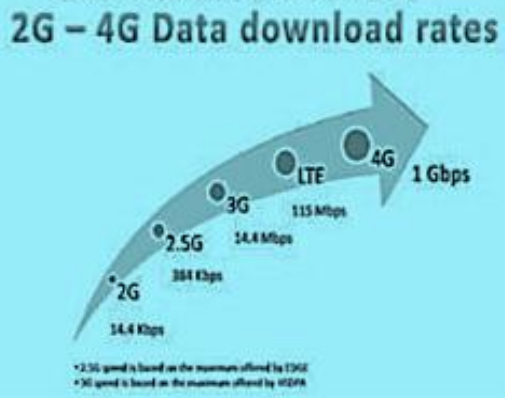
for Network Operators	for End Users
• High network throughput • Low latency • Plug & Play architecture • Low Operating Costs • All-IP network • Simplified upgrade path from 3G networks	• Faster data downloads/uploads • Improved response for applications • Improved end-user experience



Technologies of 4G includes following key features:

1. Physical layer transmission techniques
   - i. MIMO(multi-antenna and multi-user)
   - ii. Frequency-domain-equalization
   - iii. Frequency-domain statistical multiplexing
   - iv. Turbo principle error-correcting codes
2. Channel-dependent scheduling
3. Link adaption
4. Mobile IP utilized for mobility
5. IP femto cells

COMPARISON OF LTE SPEED



	3G	4G
Frequency band	1.8 - 2.5 GHz	2 - 8 GHz
Bandwidth	5 - 20 MHz	5 - 20 MHz
Data rate	Up to 2 Mbps (384 kbps deployed)	Up to 1Gbps
Access	W-CDMA	MC-CDMA or OFDM (TDMA)
Forward error correction	Convolutional rate 1/2, 1/3	Concatenated coding scheme
Switching	Circuit/packet	Packet
Mobile top speeds	200 km/h	200 km/h

Introduction to 5G System:

•A new generation of 5G standards may be introduced approximately in the early 2020s.

• 5G networks operate on rarely used radio millimeter bands in the 30 GHz to 300 GHz range.

• 5G is estimated to be 60 to 120 times faster than the average 4G latency.

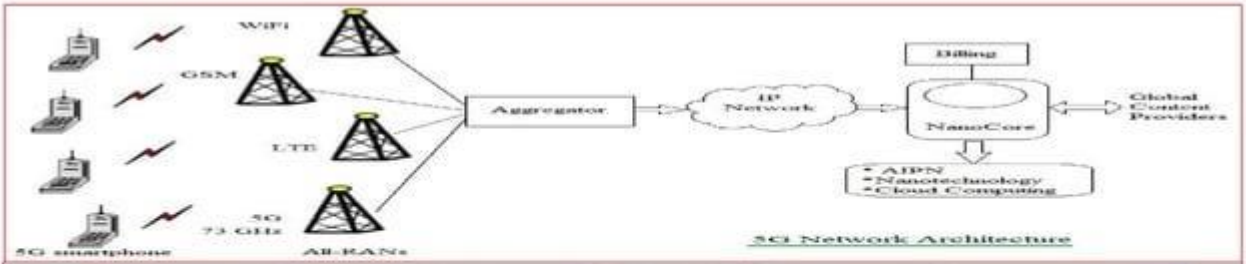
• 5G uses a scalable orthogonal frequency-division multiplexing (OFDM) framework. Testing of 5G range in mm Wave has produced results approximately 500 meters from the tower.

•Using small cells, the deployment of 5G with millimeter wave based carriers can improve overall coverage area.

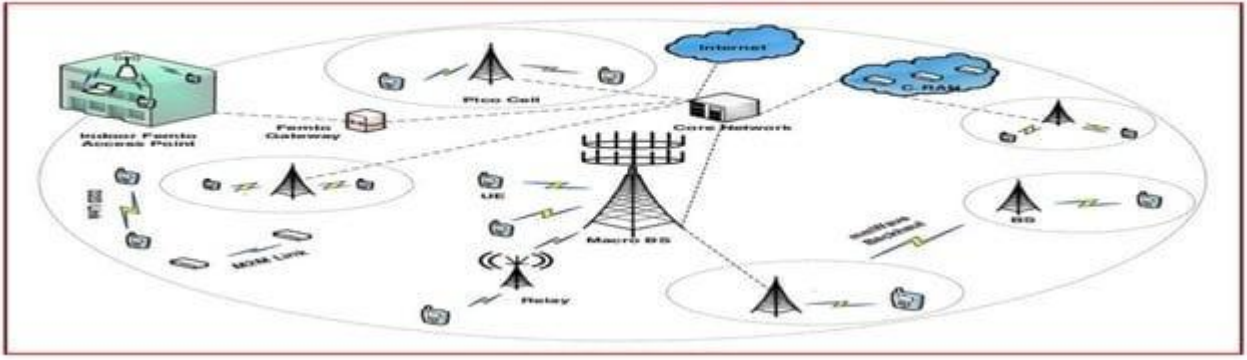
• Combined with beam forming, small cells can deliver extremely fast coverage with low latency.

According to the researches till now,
5G networks may have the following objectives:

1 Higher number of supported devices
2 Lower battery consumption
3 Lower outage probability (better coverage)
4 Higher number of simultaneously connected devices
5 Lower latencies
6 Lower infrastructure deployment costs
7 Higher versatility and scalability
8 Higher system spectral efficiency
9 High bit rates in larger portions of the coverage area


WiFi
GSM
LTE
5G
73 GHz
5G smartphone
All-RANs
Aggregator
IP Network
Billing
NanoCore
Global Content Providers
• AIPN
• Nanotechnology
• Cloud Computing
5G Network Architecture

5G CORE NETWORK
5G
RADIO ACCESS NETWORK (RAN)
USER EQUIPMENT (UE)
SMART


Internet
Pico Cell
C-RAN
Indoor Femto Access Point
Femto Gateway
Core Network
UE
Macro BS
BS
Relay

	4G	5G
Latency	60 to 98 ms	Less than 1 ms
Potential download speeds	100 Mbps – 1 Gbps	10 Gbps
Base stations	Cell towers	Small cells
OFDM encoding	20 MHz channels	100-800 MHz channels
Cell density	200-400 users per cell	100 times greater than 4G

Network	Peak speed	Average speed
5G	10 Gbps	400 Mbps
4G	1 Gbps	50 Mbps
3G	**8 Mbps**	**2 Mbps**

Comparison	2G	3G	4G	5G
Introduced in year	1993	2001	2009	2018
Technology	GSM	WCDMA	LTE, WiMAX	MIMO, mm Waves
Access system	TDMA, CDMA	CDMA	CDMA	OFDM, BDMA
Switching type	Circuit switching for voice and packet switching for data	Packet switching except for air interference	Packet switching	Packet switching
Internet service	Narrowband	Broadband	Ultra broadband	Wireless World Wide Web
Bandwidth	25 MHz	25 MHz	100 MHz	30 GHz to 300 GHz
Advantage	Multimedia features (SMS, MMS), internet access and SIM introduced	High security, international roaming	Speed, high speed handoffs, global mobility	Extremely high speeds, low latency
Applications	Voice calls, short messages	Video conferencing, mobile TV, GPS	High speed applications, mobile TV, wearable devices	High resolution video streaming, remote control of vehicles, robots, and medical procedures